

Own the Knowledge. Rent the Thinking.

Separating institutional understanding from frontier intelligence

Acropolis Technical Report 01

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Research report

Figures: [figures/acropolis-figures.html](#). Numbers come from the run reports under `../reproduction/reports/`; how they were produced is Appendix A (`methodology.md`). Section 6's cost table is from the 2026-08-20 BEAM ingest run and is marked as such.

Abstract

A company does not need to hand its institutional knowledge to a frontier model for an AI to understand it.

We find that institutional understanding and frontier reasoning can be separated: a locally operated 31B model can build the organizational record, and a frontier model can be rented only for per-question reasoning over that record. Across three public benchmarks, replacing the model that builds the organizational record with a locally operated 31B open-weight model changes downstream frontier performance by one to two points. Keeping the entire system local retains 84 to 96 percent of all-frontier performance. Renting a frontier model only for answering, where it sees the evidence for one question at a time and never the record, raises that to 96 to 99 percent.

The system that makes the separation possible is Acropolis, a governed operational model of an organization: a substrate that decides what an AI system may know and do before it does either. It holds claims rather than chunks. Every claim carries its evidence, its time of validity and its source; nothing is silently overwritten; graphs, vectors and summaries are disposable projections of a canonical record the company owns and can export. Because the record is the asset, the models around it are seats, and seats can be filled by whoever the company chooses.

We measured the substrate with one committed harness on LongMemEval-S, LoCoMo and BEAM 100K, twice: once with a frontier model in every seat, and once with every seat on a local Gemma-4-31B model. We report the retrieval, answering and governance columns for both, the negative results along the way, what is measured by construction rather than by adversarial test, and what still needs a live organization to settle.

1. Access is not understanding

An agent with credentials to the wiki, the CRM, the tracker and three years of Slack still does not know who owns a decision, which of four documents is current, what was tried last quarter and abandoned, or what it is allowed to repeat to whom.

That is the gap every enterprise AI deployment falls into, and the two standard fixes do not close it on their own.

The first fix is more context. Put the documents in the prompt, or retrieve the closest chunks and put those in the prompt, and ask the model to behave. This is what most of the market calls retrieval-augmented generation. Four things go wrong with it in practice, and none of them is intrinsic to retrieval. A model composing from fragments fills the gaps between them, and nothing in the pipeline knows a gap exists. A chunk from March and a chunk from June about the same fact are two chunks, and whichever ranks higher wins. The answer is a string, and unless something outside the model kept the citations, the evidence was a context window that no longer exists. And when access control is applied by instructing the model, the policy is enforced by the thing you are trying to constrain. Each of these can be bolted on around a retrieval pipeline: citation logging, temporal filters, contradiction checks, retrieval-time ACLs. Teams do it, one property at a time, and every one of them is a feature of that team's pipeline rather than a property of the record. Acropolis makes them properties of the substrate, so that every reader of the record inherits them by construction.

The second fix is a bigger model. It helps with the first failure and does nothing for the other three. It also carries a cost the first fix hides: to understand the company, the bigger model has to be shown the company. Every document, every thread, every customer record goes out to whoever runs the model, and what comes back is an answer, not an asset.

We took a different position. The problem is not that the model knows too little. It is that nothing in the stack holds a governed, current, attributable model of the organization in which the model can participate. Build that, and the model becomes a replaceable seat. The organizational understanding is the asset.

That position has a testable consequence, and it is the one this paper adds. If the understanding lives in a record the company owns, the model that builds the record and the model that reasons over it are different seats, and they can be filled differently. A model the company runs on its own hardware can do the understanding. A frontier model can be rented for the thinking, one question at a time, seeing only the evidence for that question. Section 2 is the experiment. The rest of the paper is what makes the result possible and what it does not yet prove.

Figure 1. The brain stays inside; a question-sized envelope can leave. Source: figures/figure-1-boundary.mmd.

```
%% Figure 1: the brain stays inside; a question-sized envelope can leave.
flowchart LR
    subgraph conventional["Conventional"]
        direction LR
        K1["Company knowledge<br/>documents, threads, records, all of it"] -->| "the whole corpus crosses the boundary" | F1["Frontier AI<br/>outside the company; understands and answers"]
        F1 --> A1["Answer<br/>an answer, not an asset"]
    end
    subgraph acropolis["Acropolis"]
        direction LR
        C2["Company<br/>observations from the systems it already runs"] --> R2["Owned organizational understanding<br/>the record: claims, evidence, time, source;<br/>built by a model the company runs"]
        R2 --> A2["Answer<br/>local answer seat: 84 to 96% of all-frontier"]
    end
```

```

R2 -->|"one question's evidence envelope, never the record"| F2["Frontier reasoning when needed<br/>96 to 99% of all-frontier"]
F2 -->|"one answer back; envelope logged with it"| A2
end
classDef inside fill:#e8f1f0,stroke:#0b6e6e,color:#1c1f24;
classDef outside fill:#f3f2ee,stroke:#8b8983,color:#1c1f24;
class C2,R2,A2 inside;
class F2,K1,F1,A1 outside;

```

2. The experiment: separating knowing from thinking

Three public benchmarks, one committed harness, no questions or conversations authored by us. LongMemEval-S is 500 questions over multi-session chat histories, 30 of them unanswerable traps. LoCoMo is 435 questions over very long two-person conversations, 88 unanswerable. BEAM 100K is 400 questions over twenty conversations of about 100,000 tokens each, across ten question types, with 40 unanswerable. BEAM is run at 100K tokens per conversation for the local campaign; the 1M variant is reported only for the frontier reference configuration in Section 5, and the two are not compared to each other in this paper. Retrieval is scored on document identity before any answering model runs; answers are judged by a fixed protocol that scores a confident wrong answer as wrong. Appendix A has the definitions, the prompts and the commands.

The system has three model seats. The understanding seat builds the record: it reads the conversations and proposes the claims the record holds. The retrieval seat rewrites the question and ranks the candidate evidence. The answer seat composes from the evidence envelope it is given, alone or in a bounded reasoning loop. Every row below fills the understanding seat with Gemma-4-31B, an open-weight model on one rented GPU, with local embeddings. "Local" throughout means ownership and operation, not the location of the machine: an open-weight checkpoint served under our own control, with no inference sent to a model-provider API. The rows differ in who fills the other two.

Figure 2. Own the knowledge, rent the thinking: how much frontier you get without a frontier brain. The two tables below are the data; "local" is Gemma-4-31B on one H200 with local embeddings, "frontier" the same frontier models as the published reference. Source: figures/figure-2-placement.mmd.

```

%% Figure 2: own the knowledge, rent the thinking. Percent of the all-frontier reference per benchmark.
%% Reference = frontier model in every seat on the published brains: LME 91.4, LoCoMo 81.8, BEAM 100K 56.2 strict / 67.5 protocol.
xychart-beta
  title "Share of the all-frontier score, by placement (percent)"
  x-axis ["LME own", "LME rent thinking", "LME rent on refusals", "LoCoMo own", "LoCoMo rent thinking", "LoCoMo rent on refusals", "BEAM own", "BEAM rent thinking", "BEAM rent on refusals", "BEAM protocol own", "BEAM protocol rent thinking"]
  y-axis "percent of all-frontier reference" 0 --> 100
  bar [95.6, 98.5, 96.9, 93.3, 96.9, 98.4, 84.0, 95.7, 86.3, 84.9, 98.4]

```

configuration	understands	thinks	LongMemEval-S	LoCoMo	BEAM 100K	BEAM 100K, own protocol
Own everything, best local configuration	local	local	87.4	76.3	47.2	57.3
Own the knowledge, rent the thinking	local	frontier answer seat	90.0	79.3	53.8	66.4
Rent only when the local answer refuses	local	local, frontier on refusals	88.6	80.5	48.5	—
Frontier in every seat, same local brains	local brains	frontier	89.6	81.8	55.2	66.8
Reference: published brains, frontier in every seat	frontier	frontier	91.4	81.8	56.2	67.5

Read as a share of the all-frontier reference, the configuration the site scores today:

	all frontier	fully local	rent the thinking	rent on refusals	questions sent out
LongMemEval-S	91.4	87.4, 95.6%	90.0, 98.5%	88.6, 96.9%	27 of 500
LoCoMo	81.8	76.3, 93.3%	79.3, 96.9%	80.5, 98.4%	40 of 435
BEAM 100K, strict	56.2	47.2, 84.0%	53.8, 95.7%	48.5, 86.3%	8 of 400
BEAM 100K, own protocol	67.5	57.3, 84.9%	66.4, 98.4%		

- Every row in both tables has full per-question reports in the published bundle (Appendix A.12).

"Own everything" is the better of the local single-call and local-loop configurations, with no frontier model anywhere: the single call on LongMemEval (87.4; the loop reads 86.0), the loop on LoCoMo and BEAM (the single call reads 73.6 and 42.0). Renting the thinking keeps the record, the retrieval and the ranking local and rents one seat, the answer; the frontier model sees one question's evidence envelope, a few thousand tokens of claims and excerpts, and the envelope is stored with the answer. Renting on refusals keeps the local loop's answer wherever it answered, and sends out only the questions it declined. With the local single-call answers as the base instead, LongMemEval reads 88.4 or 88.8 depending on which frontier row answers the refusals. BEAM is scored two ways because strict correctness gives no credit for a partially ordered list or a partial summary; BEAM's own protocol does, and the appendix prints both.

Three things are in those tables, and they are the argument of this paper.

The understanding seat holds. Brains a local model built, with about 60 percent as many accepted claims as the frontier extractor produced on the same conversations, score within one to two points of the published brains once a frontier model runs the other seats: 89.6 against 91.4, 81.8 against 81.8, 55.2 against 56.2. Retrieval on them is the same or better: Hit@5 99.2 against 99.0, 91.2 against 90.7, 39.2 against 40.6. The material gap does not appear in what the local model writes down; it arises downstream, in the seats that read it.

The cost of owning everything is in the reading seats, and it grows with the length of what has to be read. The best local configuration trails frontier-everywhere on the same brains by 2.2 points on LongMemEval, 5.5 on LoCoMo and 8.0 on BEAM. On every benchmark the loss has the same

shape: the local model ranks candidates less well over long excerpts (Hit@5 80 against 91 on LoCoMo, 34 against 39 on BEAM, on identical pools), and it composes less well from them. A cross-encoder reranker was tried as the cheap fix and refuted: on the same LoCoMo pool it ranked 28 points below the local language model, because ranking a long excerpt against a question is a reading job. A contract rewrite aimed at the local answer seat's observed mistakes was tried on both LoCoMo conversations and lost half a point. The loop is not a free lever either: it pays 2.7 points on LoCoMo and 5.2 on BEAM, and costs 1.4 on LongMemEval. The gap is model capacity at length, and the paper says so rather than dressing it.

Renting the thinking buys most of it back. Swapping only the answer seat for a frontier call lifts the local stack to 90.0, 79.3 and 53.8, which is 98.5, 96.9 and 95.7 percent of the all-frontier reference. On BEAM's own protocol that row, 66.4, is 98.4 percent of it and within noise of frontier everywhere, 66.8. On LongMemEval it is above the site's published core number. Renting only on refusals sends 2 to 9 percent of questions out and reads 97 and 98 percent on the two benchmarks where the local model refuses often enough for it to matter; the two mechanisms together, local retrieval with a rented answer and escalation on its refusals, read 91.4 on LongMemEval, the reference itself, with 21 questions escalated.

Figure 3. BEAM 100K by ability: what renting the answer seat buys back. Series in order: own everything (local loop), rent the thinking, frontier in every seat on the same brains. Overall 57.3, 66.4, 66.8. Source: figures/figure-3-beam100k-by-ability.mmd.

```
%% Figure 3: BEAM 100K by ability under BEAM's own protocol (mean rubric score of 40 q
uestions per ability).
%% Three placements on the same local brains: own everything (local loop), rent the th
inking (frontier answer seat), frontier in every seat.
xychart-beta
    title "BEAM 100K by ability: local loop vs rented thinking vs frontier, same brain
s"
    x-axis ["extraction", "preference", "instruction", "update", "contradiction", "tem
poral", "multi-session", "summary", "ordering"]
    y-axis "mean rubric score (percent)" 0 --> 100
    line [70.8, 94.1, 72.8, 59.7, 52.7, 66.7, 60.3, 23.8, 16.5]
    line [89.1, 87.8, 90.9, 57.9, 67.5, 78.6, 58.0, 36.9, 32.9]
    line [86.4, 87.9, 92.1, 62.5, 67.5, 80.7, 55.1, 36.3, 34.6]
```

The guard columns survive the swap. With no frontier model in any seat, 29 of the 30 LongMemEval traps are refused. On LoCoMo the local rows refuse 70 to 75 of 88 traps against 70 to 80 for the frontier rows. Provenance is the same in every row because it is a property of the record, not of the model reading it.

Two caveats on the BEAM table. Event ordering and summarization read 0 to 3 of 40 for every configuration under strict correctness and 15 to 37 under the protocol, local or frontier; they are the benchmark's hardest types for everyone. And contradiction resolution, the question type closest to this paper's thesis, is under half for every row: the answers name the disagreement and quote both sides, as the contract instructs, but the two evidence messages reach the envelope together on too few questions. That is the reranker again.

The model is a replaceable seat. The organizational understanding is the asset. The rest of the paper is about the record that makes that true.

3. What Acropolis is

One sentence carries the topology: Varys sees, Parthenon holds, Eunomia orders, Hero acts and communicates, Pythia answers, Ariadne reaches, MUXI runs it all. Acropolis is the machine.

The names are deliberate. People-names act; place-names hold. Nothing that holds may act, and nothing that acts may hold. That rule is the architecture.

Parthenon holds. It is the brain: a client-owned, versioned, exportable store of observations, evidence, claims and the projections built from them. It never acts. It receives proposals through one signed intake door, validates them, and accepts, rejects, deduplicates or supersedes. A claim in Parthenon says: source S stated X at time T, here is the evidence span, here is the interval over which X was true, here is what superseded it and when. Nothing is deleted; things are corrected, disputed or superseded, and the change log records which.

Varys sees. Connectors, backfill, triage, extraction, proposal. Varys reads the approved systems, distils observations into candidate claims and submits them to Parthenon's door. It never speaks to a person and never changes operational state. Everything it proposes can be refused. The model that does the distilling is the understanding seat of Section 2.

Eunomia orders. The identity service. It fronts the client's own directory and answers one question for both Hero and Parthenon: who is this, what groups, whose data. It stores no memberships of its own. The directory stays the single source of identity truth.

Hero acts and communicates. The agent harness, built on MUXI, the open-source agent-formation substrate. Hero lives in Slack or Teams, runs formations, carries the channels. One Hero binds to one brain by structure, not by filter: a work Hero reads the company Parthenon, a personal Hero reads a private one.

Pythia answers. Some questions need working out rather than looking up: how long between two events, how many devices in total, whether the absence of a fact is an answer. Pythia is a bounded loop with three tools, ask Parthenon, compute deterministically, extract, and a mandatory time anchor. No world tools, ever. Her answers carry a second kind of warrant: derived from claims X and Y by operation O, beside Parthenon's asserted by source S at time T. The model inside the loop is the answer seat, the one seat Section 2 shows is worth renting.

Ariadne reaches. A desktop agent, paired per user with consent on the user's machine, dispatched by Hero when the work is local. She grants no authority and holds no memory.

Put those together and the ordinary agent request changes shape. Instead of retrieve, stuff, hope, it becomes: resolve identity and policy, retrieve under that authorization, assemble task-scoped context with an explicit coverage state, plan, act with permission, record an attributable outcome, update versioned state. Observation is read-only. Memory is persistent. Execution is permissioned. Identity travels with the request. Action happens at the edge.

Figure 4. Bolted-on retrieval and the resident request path, step by step. Each failure mode sits where it enters the first path; each Acropolis step names the component that decides it. Source: figures/figure-4-request-path.mmd.

```
%% Figure 4: bolted-on retrieval vs the resident request path, step by step.
flowchart LR
```

```

subgraph bolted["Bolted-on: retrieve, stuff, hope"]
  direction LR
  B1["Retrieve nearest chunks<br/><i>stale wins if it ranks higher</i>"] --> B2["Put
them in the prompt<br/><i>model sees what it should not</i>"] --> B3["Instruct the mod
el to behave<br/><i>policy enforced by the thing being constrained</i>"] --> B4["Answ
er<br/><i>gaps filled; no provenance survives</i>"]
end
subgraph resident["Resident: the Acropolis request path"]
  direction LR
  S1["Resolve identity and policy<br/>Eunomia, from the client's directory"] --> S2
["Retrieve under that authorization<br/>Parthenon, before any model runs"] --> S3["Ass
emble task-scoped context<br/>explicit coverage state"] --> S4["Plan, act with permiss
ion<br/>Hero; Pythia when working out is needed"] --> S5["Record an attributable outco
me<br/>update versioned state"]
end
classDef acr fill:#e8f1f0,stroke:#0b6e6e,color:#1c1f24;
class S1,S2,S3,S4,S5 acr;

```

This is the architecture for resident AI: the AI lives inside the company that owns it, runs where the work already happens, sees the whole operation, runs on infrastructure the company owns, and what it learns belongs to the company in a form it can export. The full thesis is at varops.com/resident-ai. A resident is not a tool invoked per query. It accumulates context, carries identity, and needs what a new employee needs on day one: to know what is current, who owns what, and what it is allowed to do. The fourth condition is why Parthenon is client-owned and exportable rather than a service the company rents.

The threat model is explicit. Acropolis is built to prevent an over-privileged agent reading what its principal may not; leakage across brains or tenants; and stale or contradictory context leading to a confidently wrong action. It does not address a compromised endpoint, an insider who legitimately holds high privilege, or a compromised model provider. When an external model is rented for a seat, the evidence envelope for that question leaves the customer's perimeter: a few thousand tokens of claims and excerpts, never the corpus and never the record. The envelope is logged with the answer, so what leaves is itself auditable. Section 2 quantifies how often that happens under refusal escalation, 2 to 9 percent of questions, and what it buys. Only a fully local deployment sends nothing out, and Section 2 reports what that costs.

4. Seven properties, and what each one measures

The properties below are not delegated to prompt compliance. Retrieval, authorization, coverage accounting and provenance are code paths and record structure; the model composes inside the boundary they set. Some benchmark performance does depend on how the model is instructed, and the paper measures that too. Here is each property, with its measurement where one exists and the plain words "by construction" where a test has not attacked it.

What the model sees is decided before it runs. Retrieval, authorization and coverage accounting are deterministic code paths that run before any model is called. The model composes an answer from an evidence envelope; it never decides what it was allowed to see. Whether it asserts beyond that envelope is a contract instruction, and its compliance is what the guard columns measure: across four full LongMemEval runs, 25 to 28 of the 30 unanswerable-trap questions were correctly

refused each run, and one answer in the 120 trap rows was classified as a fabrication under manual review. With no frontier model in any seat, 29 of 30. That is one fabrication in the tested trap sample under the stated protocol; it is not a general hallucination rate, and the paper does not treat it as one.

Absence is labeled, never silent. Every question resolves to an explicit coverage state: answered, searched-and-empty, or outside-the-ontology. "The system found nothing" and "the system holds no concept of this" are different answers and callers can tell them apart. This is why an abstention column exists at all: BEAM abstention 75.7 under the governed composite, against 52.5 for Vendor X, the commercial memory layer of Section 5. We also measured what the column costs, by removing it. The same LongMemEval configuration instructed never to say the information is missing scored 87.2 against 88.0 with the refusal intact, and turned 30 refusals into 30 fabrications. Scored on answerable questions only, the way never-refuse systems report, the same answers read 92.8. The refusal buys nothing on the headline and everything on the guard column.

Every answer carries its evidence. Answers cite the claims they rest on; each claim carries its source, timestamp and evidence span; derived conclusions carry the trace of what they were computed from. An auditor can walk any answer back to the conversation line it came from. Measured exhaustively on the stored runs: all 118,890 claims delivered across 2,500 LongMemEval answers resolve to a stored observation with a source identity, and an audit of 2,233 answers against the evidence each received finds unsupported wrong assertions in about one answer in a hundred, at the same rate for local and frontier models.

Access control at the substrate, not in the prompt. Callers hold cryptographic identities; every request carries signed authorization; scopes resolve at the substrate itself, per caller. A caller below a scope cannot distinguish "withheld" from "empty", so permissions cannot be probed through absence. Isolation is one Postgres schema per brain, failing closed. It has been attacked: an adversarial probe against live per-brain servers, five brain pairs and nine cases each, refused or contained all 45. Wrong-tenant, foreign-key, tampered, expired and missing assertions are refused with no data; an assertion replayed against another brain's server returns none of the first brain's claims; naming another brain or another principal's scope is refused.

One writer, one validation path. Nothing a model says grants itself storage. All writes flow through the signed intake door, which validates, deduplicates, supersedes and can reject. On the BEAM ingest a first conversion emitted human-readable dates instead of RFC 3339; the door quarantined 5,732 submissions and accepted zero claims rather than store them. After the fix: 388 submitted, 388 accepted. A store would have indexed the garbage.

Reproducible by construction. An answer records the snapshot, plan and authorization decision it was produced under. Every benchmark report records the dataset hash, the code revisions with a dirty-tree flag, and the model pins. The reports behind every number in this paper are published with it, under the release gate in Appendix A.12.

Figure 5. Refusal posture is a dial with two measured endpoints. Moving from synthesis-forward to balanced buys 14.3 points of BEAM abstention for 1.1 points of LongMemEval accuracy. Source: figures/figure-5-refusal-dial.mmd.

```
%% Figure 5: refusal posture is a dial with two measured endpoints (one contract, two
calibrations).
%% synthesis-forward: LongMemEval 90.5, BEAM abstention 50.0. balanced: LongMemEval 8
```

```
9.4, BEAM abstention 64.3.
```

```
xychart-beta
```

```
  title "Refusal posture: what each calibration costs and buys"
```

```
  x-axis ["synthesis-forward LME", "synthesis-forward BEAM abstention", "balanced LM  
E", "balanced BEAM abstention"]
```

```
  y-axis "percent" 0 --> 100
```

```
  bar [90.5, 50.0, 89.4, 64.3]
```

Governance is a dial, not a destiny. Refusal posture is a deployment configuration with two measured endpoints. The synthesis-forward setting scores LongMemEval 90.5 and BEAM abstention 50.0; the balanced setting scores 89.4 and 64.3. One contract cannot maximize both, and we measured that rather than asserting it: three contract variants scored 56.4, 57.4 and 57.1 overall on BEAM. Knobs redistribute points between refusal and coverage. They do not create them. We confirmed it once more in September: a fourth contract written against the local model's observed mistakes moved LoCoMo from 73.0 to 72.2 on 435 paired questions, gaining on some and losing on others, with the refusal count unchanged. Systems that weld in a never-refuse posture cannot publish this column, and do not.

Table 1. Evidence map: every property, with its status. Measured: a run exercised the property and a number depends on it. By construction: built in, never attacked. Specified: designed, no benchmark exercises it. Source: `figures/table-1-evidence-map.md`.

property or path	status	evidence
Signed intake, distillation, claims with evidence and supersession	measured	Exercised by every benchmark run; door quarantined 5,732 malformed submissions on the first BEAM conversion
Retrieval at published budgets, identity-scored	measured	Figure 6; Hit10 99.80 / 93.06 / 44.00. Unchanged when the extractor is swapped: on brains built by a local 31B model, Hit5 99.2 / 91.2 / 40.6 against 99.0 / 90.7 / 40.6 on the published brains (LongMemEval, LoCoMo, BEAM 100K)
Absence labeled: coverage states in the envelope	measured	BEAM abstention 75.7 vs 52.5; 25 to 28 of 30 traps refused on LongMemEval; 29 of 30 with no frontier model in any seat. Never-refuse measured on the same brains: removing the refusal moved LongMemEval 88.0 to 87.2 and turned 30 refusals into 30 fabrications
Pythia loop: ask, compute, extract with a time anchor	measured	Agentic row 90.5 vs core 89.2, four full runs
Governed composite routing on refusal and contradiction signals	measured	Figure 8; validated live before the full run
Refusal posture as a configuration	measured	Figure 5; two endpoints, three variants at 56.4 / 57.4 / 57.1; escalation on the system's own refusals measured at 88.4 to 91.4 across four local configurations
Scores do not depend on the judging model	measured	A local 31B judge agrees with the frontier judge on 98.7 to 99.2% of 2,223 LongMemEval answers and preserves every ordering; its disagreements are stricter, protocol-literal calls
Per-brain isolation, fails closed	measured	Adversarial probe on live per-brain servers, five brain pairs, nine cases each, 45 of 45 refused or contained: wrong tenant, foreign key, tampered, expired and missing assertions refused with no data; an assertion replayed against another brain's server returns none of the first brain's claims; naming another brain or another principal's scope is refused
Every answer carries its evidence	measured	All 118,890 claims delivered across 2,500 stored LongMemEval answers resolve to a stored observation with a source identity, 0 exceptions; an LLM audit of 2,233 answers against their delivered evidence finds unsupported wrong assertions in about 1 in 100 answers, evenly across local and frontier rows
Graduated disclosure	measured	Rungs derived at write time, never redacted at read time; 18 integration tests including an adversarial reconstruction over 22 reader-visible tables. End to end over the wire: a finance-group reader's envelope carries the precise value 2,140,000; a sales-rung reader's envelope on the same subject carries the band 2.0M to 2.5M and the trend "up", no byte of either precise value and no parent claim id
Task-scoped context: ask_batch, context(entities), validity token	measured	Built and drilled over the wire: context over several subjects returns one envelope with a section per subject; ask_batch answers several questions under one assertion, one scope and one snapshot; every envelope carries a validity token, and revalidate reports it stale after the first lifecycle event in its scopes (measured: 3 events after one supersession) while a token for another brain or an ungranted scope is refused. No public benchmark exercises the two-stage ask; the measurement is the mechanism drill
Identity resolution and permission-	measured	Resolver: 1,000 people across 3 namespaces with 50 evidence-backed duplicate merges and 20 conflicting handles: 3,030 correct, 0 wrong, 0 missing,

property or path	status	evidence
change propagation, bounded fixture		the 20 conflicts surfaced as typed ambiguity naming both candidates, duplicates follow to the primary, a reversed merge restores the source, ~2 ms per resolve. Propagation: a directory change reaches the next issued assertion in 0 ms with the cache off and within the cache TTL with it on (measured 2.06 s at a 2 s TTL). With the generation feed on, an assertion already in a caller's hand is refused within the poll interval (measured 976 ms at a 1 s poll, 3 s by default) instead of at its expiry; a feed outage is honoured through a grace window and then fails closed (measured 2,993 ms at 3 s). Live directories, aliases at scale and reorganizations are not exercised
Sovereignty tier: open-weight models on sub-\$100k hardware	measured	Figure 2. Every model seat on Gemma-4-31B, one H200, on all three benchmarks: fully local 87.4 / 76.3 / 47.2 (LongMemEval, LoCoMo, BEAM 100K strict; best local configuration per benchmark); local knowledge with the frontier answer seat 90.0 / 79.3 / 53.8; frontier in every seat on the same local brains 89.6 / 81.8 / 55.2. Brains a local model built score within one to two points of the published brains under frontier seats on every benchmark

5. The frontier configuration, and the comparison

Section 2 reported the local campaign against the all-frontier reference. This section is the reference itself, the configuration the site has published since August: a frontier extractor, a frontier reranker, a frontier answer. Everything is identity-scored at the published budgets.

Figure 6. Retrieval holds to a million tokens, then meets the cliff every published figure shows.

First series Hit@5, second Hit@10; the first three points are the frontier configuration, the rest the September local-brain rows. Source: figures/figure-6-retrieval.mmd.

```
%% Figure 6: retrieval holds to a million tokens, then meets the cliff every published
figure shows. Hit@5 and Hit@10, identity-scored.
xychart-beta
    title "Retrieval: Hit@5 (first series) and Hit@10 (second series)"
    x-axis ["LME frontier", "LoCoMo frontier", "BEAM 1M frontier", "LME local brains,
local rerank", "LoCoMo local brains, local rerank", "LoCoMo local brains, frontier rer
ank", "BEAM 100K published brains", "BEAM 100K local brains, frontier rerank", "BEAM 1
00K local brains, local rerank"]
    y-axis "percent of questions" 0 --> 100
    line [99.0, 90.74, 35.84, 99.4, 80.6, 91.2, 40.6, 39.2, 33.8]
    line [99.8, 93.06, 44.0, 99.6, 85.2, 92.6, 49.6, 47.3, 43.9]
```

Retrieval

	LongMemEval-S	LoCoMo	BEAM 1M
questions	500	432	700
corpus per question	multi-session history	10 long conversations	~1M tokens
Hit@10	99.80%	93.06%	44.00%
Hit@5	99.00%	90.74%	35.84%
Recall@10 (strict)	95.80%	82.64%	16.16%
Recall@10 (micro)	97.36%	77.29%	8.86%
MRR	0.9846	0.8225	0.2546

● August retrieval reports partially retained; the local-brain retrieval rows in Figure 6 are ●.

Read the BEAM column first if scale is the question. Every system we have measured, or for which comparable figures are published, falls off a cliff at a million tokens per question, ours included, and the memory-layer comparison's own figures show the same cliff: 94.4 on LongMemEval, 64.1 on BEAM 1M, 48.6 on BEAM 10M. We publish the retrieval figures at that scale. Most competitors do not. Retrieval is identical across every answering configuration below; the agentic layer consumes the same ranked evidence the direct path does.

Figure 7. LongMemEval answering: the published configurations, the local-brain row, the never-refuse probe, and a comparison that cannot lose points for refusing. Source: figures/figure-7-longmemeval-answering.mmd.

```
%% Figure 7: LongMemEval answering, published configurations, the local-brain rows, the
never-refuse probe, and the comparison that cannot lose points for refusing.
xychart-beta
    title "LongMemEval-S accuracy, 500 questions"
    x-axis ["core", "agentic (Pythia)", "agentic + escalation", "local brains, rented
answer seat", "never-refuse probe, all 500", "never-refuse probe, 470 answerable only", "Vendor X, never-refuse protocol"]
    y-axis "accuracy (percent)" 0 --> 100
    bar [89.2, 90.5, 91.6, 90.0, 87.2, 92.8, 98.0]
```

Answering, LongMemEval-S

configuration	accuracy	notes
Acropolis core	89.2%	band 88.4 to 89.6 over repeated full runs
Acropolis agentic (Pythia)	90.5%	band 90.4 to 90.6, four full runs
Acropolis agentic + frontier escalation	91.6%	escalation only on the system's own refusals
Vendor X ¹	98.0%	never-refuse protocol

● The three Acropolis rows are re-runnable from the tagged harness; their August per-question reports are partially retained (Appendix A.12). The local-brain rows of Figure 7 and the reference rows on the published brains are ●.

Vendor X is a commercial memory layer: retrieval and answering over conversation history, with none of the identity, authorization, or execution components described in Section 3. Its figure is

under a never-refuse protocol and is not directly comparable to Acropolis's accuracy: by design, that protocol cannot lose points for refusing. We measured the size of that bias on our own system rather than estimating it. Under a never-refuse contract, the same brains, retrieval and judge score 87.2 on all 500 questions and 92.8 on the 470 answerable ones (the 30 traps count as wrong when not refused). Because 30 of the 500 questions are unanswerable traps that count as correct only when refused, a system that never refuses caps at 94.0 under the official rule. A published 98.0 therefore must score those traps differently or exclude them, and it is judged by a protocol that does not treat a confident wrong answer on an unanswerable question as wrong. In practice, the 98.0 cannot be compared to Acropolis's 89.2 to 91.6 as if they were measured under the same rules.

Governance columns: 25 to 28 of 30 unanswerable traps correctly refused per run; one fabrication in the tested trap sample under the stated protocol.

Answering, LoCoMo

Accuracy 81.8% on the 432-question sample the site publishes, Hit@5 90.7.

Figure 8. BEAM 1M per category: where the governed composite leads and where it trails. First series the governed composite, second Vendor X as published. Source: `figures/figure-8-beam1m-composite.mmd`.

```
%% Figure 8: BEAM 1M per category, governed composite vs Vendor X (published), official rubric protocol.
xychart-beta
  title "BEAM 1M: governed composite (first series) vs Vendor X (second series)"
  x-axis ["abstention", "contradiction", "summarization", "multi-session"]
  y-axis "score (percent)" 0 --> 100
  line [75.7, 62.1, 56.4, 57.9]
  line [52.5, 35.7, 63.5, 65.2]
```

Answering, BEAM 1M (full 700 questions)

configuration	overall	abstention	contradiction	summarization	multi-session
Acropolis core ²	60.2	74.3	64.7	31.9	58.6
Acropolis governed composite ³	+3.5 over its paired core	75.7	62.1	56.4	57.9
Vendor X ¹	64.1	52.5	35.7	63.5	65.2

● Re-runnable from the tagged harness; August per-question reports partially retained (Appendix A.12).

The governed composite is a deterministic routing policy over two complete runs: a cheap direct pass's refusals and contradiction declarations ship as they are, 150 of 700 rows, and a frontier synthesis loop answers the rest. No oracle, no picking the better score per question. Its headline is abstention 75.7, the highest of any configuration we have measured and 23 points above the comparison, with contradiction handling 26 points above, while nearly doubling the core system's

summarization. It lands within about three points of the comparison's overall figure under a cross-protocol comparison, with every refusal intact.

Full core per-category: preference 83.2, extraction 82.1, instruction 77.8, abstention 74.3, contradiction 64.7, temporal 63.5, knowledge-update 63.4, multi-session 58.6, summarization 31.9, event-ordering 26.0.

Table 2. Head to head, with the protocol differences on the table. Source: figures/table-2-head-to-head.md.

evaluation	Acropolis core	Acropolis agentic	Acropolis composite	Vendor X ¹
LongMemEval-S accuracy	89.2	90.5	—	98.0
LongMemEval-S unanswerable traps refused (of 30)	—	25 to 28	—	—
LongMemEval-S Hit10	99.80	99.80	—	—
LoCoMo accuracy	81.8	—	—	92.5
BEAM 1M overall (official protocol)	60.2 ²	—	+3.5 paired ³	64.1
BEAM 1M abstention	74.3	—	75.7	52.5
BEAM 1M contradiction resolution	64.7	—	62.1	35.7
BEAM 1M summarization	31.9	—	56.4	63.5
BEAM 1M multi-session reasoning	58.6	—	57.9	65.2
BEAM 1M Hit10	44.00	44.00	44.00	—
Publishes retrieval at 1M tokens	yes	yes	yes	—
Publishes an abstention column	yes	yes	yes	BEAM only

¹ Vendor X figures as published by the vendor under its own protocol, which instructs the model never to state that information is missing, so a refusal can never cost it a point; it does not publish retrieval or abstention metrics at the 1M-token scale. It is a memory layer only: no identity, authorization, or execution components. Acropolis figures include honest refusals and are measured under our documented aggregation; both sides' guard columns are shown so the trade is visible.

² Core band from the August gate campaign under its published aggregation.

³ Governed composite measured on the full 700 questions as a deterministic routing policy over two complete runs, validated live on the governance categories. Its overall is stated as a paired delta because it was measured under a rebuilt aggregation formula; absolute cross-formula comparisons are not valid, per-column and paired figures are.

6. The cost of record

A lossless index and a governed, intentionally lossy, multi-tenant substrate on rented CPU are not the same kind of thing, and the comparison above should be read knowing what the second one

costs to run. These figures are from the BEAM 1M ingest of 2026-08-20: 35 conversations, roughly 35 million tokens.

Figure 9. The cost of record. Source: figures/figure-9-cost-of-record.mmd.

```
%% Figure 9: the cost of record. BEAM 1M ingest of 2026-08-20: 35 conversations, about
35 million tokens, two Xeon 8280s, 31 GB, no GPU.
flowchart LR
    I["Ingest, one time<br/><b>$8.24</b><br/>17,745 accepted claims · 37,387 LLM calls"]
    --> C["Per accepted claim<br/><b>$0.0021</b><br/>about $0.24 per million-token convers
ation"]
    C --> P["Index, one time<br/><b>$0</b><br/>72,986 documents embedded, local nomic, C
PU only"]
    P --> Q["Retrieval, per question<br/><b>$0.0037</b><br/>$2.28 for 625 questions"]
    I --> D["Quarantined at the door<br/><b>0</b><br/>after the RFC 3339 fix; 5,732 refu
sed before it"]
    classDef tile fill:#f7f6f2,stroke:#dcd9d0,color:#1c1f24;
    class I,C,P,Q,D tile;
```

stage	measured
hardware	2 × Intel Xeon Platinum 8280, 16 cores, 31 GB, no GPU
ingest, one time	17,745 accepted claims for \$8.24 over 37,387 LLM calls
	\$0.0021 per claim, about \$0.24 per million-token conversation
	0 quarantined: the door accepted nothing malformed
projection, one time	72,986 documents embedded for \$0, CPU only, local nomic
retrieval	\$0.00365 per question

Two facts most stores cannot claim. The embedder is free and local: there is no embedding vendor in this stack and no GPU in these machines. And the whole corpus cost under ten dollars to ingest and nothing to index.

Distillation is deliberately lossy. We throw information away on the way in and still put the right evidence in the top ten on 99.8% of LongMemEval questions. Compression that keeps the answer is a system property, not a retrieval trick.

The local campaign adds the other half of the bill. Building the LoCoMo brains took 53 and 70 minutes per conversation on one H200 with 16 extractions in flight. Answering is where a 31B model is slow: on BEAM 100K the fully local rows ran at a median of 12 to 15 minutes per question with 14 conversations sharing the GPU, against under a minute for the frontier rows, because the synthesis prompts run near 57,000 tokens. That is a throughput figure for a shared benchmark box, not a single-user latency, and it is the honest price of the "own everything" row.

What this means for a deployment

The benchmarked layer is the knowledge substrate: ingestion, distillation, retrieval and answering over conversational history. The governance properties are architectural: authorization before context, explicit absence states, provenance on every answer. The tables show they coexist with strong retrieval and answering on public datasets, and that the understanding half of the system can run on hardware the company owns without moving the numbers: 84 to 96 percent of the all-

frontier score with nothing rented, 96 to 99 percent with the answer seat rented. That is the sovereignty claim in concrete form: the company can own the brain and keep most of the performance. What is measured on fixtures rather than live organizations are the enterprise-hardening properties: identity resolution at scale, permission propagation through a real directory, and operational outcomes. Those are the next validation steps, not assumed capabilities; a buyer should treat them as such when pricing risk and rollout.

7. Limitations

7.1 Measured on fixtures, not live organizations

All identity, permission and task-scoped results in this paper are measured on controlled fixtures: a generated organization, a synthetic directory, scripted changes. They demonstrate mechanism and correctness under test conditions, not behavior under a real company's directory, reorganizations and operational load. Section 7.2 details the fixtures; Section 8 lists the live-organization tests that would move these from mechanism demonstrated on fixtures to validated in production.

7.2 Identity, permission and task-scoped fixtures

Identity resolution and permission-change propagation were measured on a bounded fixture rather than a live directory: a generated organization of 1,000 people across three namespaces, with 50 evidence-backed duplicate merges and 20 conflicting handles. The fixture resolved 3,030 references with none wrong or missing, surfaced each conflict as typed ambiguity naming both candidates, and restored a reversed merge; a directory change reached the next issued assertion immediately with the cache off and within the cache TTL with it on. An assertion already issued carries the directory generation it was issued against, and a deployment that polls the generation feed refuses it within the poll interval once the directory changes: measured at 976 milliseconds with a one-second poll, three seconds by default, against a five-minute assertion. When the feed is unreachable the last known generation is honoured through a grace window and the deployment then fails closed, which was measured too. Aliases at scale, reorganizations and a live directory remain unexercised.

Task-scoped context, the two-stage path where an agent asks for constraints before planning and for entity context after candidates exist, is built and drilled rather than benchmarked: a batch of questions is answered under one assertion, one scope and one snapshot; context is returned for several entities in one envelope; and every envelope carries a validity token that a revalidate call reports stale after the first change in its scopes, while a token for another brain or an ungranted scope is refused. No public benchmark exercises the two-stage ask, so its row is measured by mechanism, not by score.

Graduated disclosure is built and measured: rungs are derived when a fact is accepted, superseded or retracted, never redacted at read time, and an end-to-end drill shows a finance reader receiving the precise value while a sales reader receives the band and the trend on the same subject with no byte of the precise value in the envelope. What is not claimed is coverage of the rule kinds beyond band, trend and the floors the tests exercise.

7.3 What is not claimed

The local placement was measured on one open-weight model, Gemma-4-31B, on one rented H200. No claim is made about other open-weight models or smaller hardware; the sub-\$100,000 figure is a hardware class, not a measured bill. The BEAM local rows are on the 100K variant, not the 1M variant the site publishes for the frontier configuration, and the two are not compared to each other anywhere in this paper. The 1M variant on a local model is a cost question, not an open one, and it was not run.

The judge is a language model operating under a fixed prompt. A local 31B model re-judging 2,223 stored LongMemEval answers under the same prompt agreed with it on 98.7 to 99.2 percent of verdicts and preserved every ordering; on LoCoMo, where gold answers are longer free-text spans, agreement was 95.2 percent and the local judge was stricter on every row. The scores do not hinge on which model judges. The one human step in scoring is the manual classification of the non-refused trap rows, which changes no score and is published for re-classification. Comparisons to other systems are cross-protocol and labeled so wherever they appear. None of the three datasets is enterprise-shaped.

Two negative results are reported rather than buried. A local cross-encoder reranker did not close the local retrieval gap; it widened it. And prompt contracts written against observed failure modes did not move scores: three variants on BEAM in August landed within a point of each other, and one written for the local model in September lost half a point on 435 paired questions.

8. Posture, and what comes next

Refusal posture ships as a labeled configuration. A deployment picks the balanced setting or the synthesis-forward one knowing what each costs on each benchmark, and the paper reports both endpoints rather than the more flattering one.

Model placement ships the same way. A deployment picks own everything, own the knowledge and rent the thinking, or rent only on refusal, and Figure 2 is the price list. The recommended default is the middle row: the company builds and holds the record on hardware it owns, and rents a frontier model only for per-question answering, which sees only that question's evidence envelope. On the benchmarks this configuration scores 96 to 99 percent of the all-frontier reference, exceeds the published core number on the easiest dataset, and keeps the organizational record on hardware the company owns. A deployment that wants nothing rented at all starts at 84 to 96 percent and can escalate its own refusals, 2 to 9 percent of questions, to close most of the rest.

The model is a replaceable seat, and we have the receipt three times over: swapping the extraction model from a frontier vendor to a 31B open-weight model moved the frontier-seat scores by one to two points on every benchmark, measured on identical questions. The organizational understanding is the asset. It is the part that does not change when the model does: provenance, state, identity, policy, boundaries.

What is next is the reading seat: a local model that ranks and composes over long excerpts as well as a frontier model does. That is the one gap the local campaign left, and the levers that pay there are a stronger open-weight reranker and better local retrieval, not another contract clause. The tests in Appendix A.14 that still need a live organization, identity at real scale, permission changes through a real directory, operational outcomes, are the other campaign, and the paper will not claim them until one has run.

The harness, the aggregator, the per-question reports and the manifest that ties them to the figures above are published with this paper. Re-run what we ran. Where a figure cannot be independently re-run, the paper says so next to the figure.

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